

Table of Contents

Cisco Secure Workload — Australian Cyber Security Centre (ACSC) Essential Eight

Reference Design | Essential Eight Maturity Model (EEMM)

Version: 1.0

Use Case: Fresh Install, Hybrid Environment (On-Prem + Cloud)

Reference framework: ACSC *Essential Eight Maturity Model* (strategies E1–E8; Maturity Levels ML1–ML3)

Reader's Guide

Who this is for. Australian Government entities and organizations aligning with the ACSC Essential Eight; security architects translating the eight mitigation strategies into technical evidence; and assurance teams preparing for assessments against ML1–ML3 while using Cisco Secure Workload (CSW) as part of the control stack.

Questions this reference design helps you answer:

- Where does CSW provide **primary** technical evidence versus **complementary** visibility alongside dedicated tools (application control, patching, MFA, backups)?
- How do I map CSW telemetry and segmentation to each Essential Eight strategy (E1–E8)?
- What can I demonstrate at **ML1**, **ML2**, and **ML3** maturity levels using CSW exports and enforcement artifacts—without over-claiming coverage for strategies that are fundamentally endpoint- or identity-provider-led?
- How do I prioritize patch effort using CVE, EPSS, and reachability context (**E2**, **E6**)?
- How do I evidence restriction of administrative privilege **paths** and visibility into privileged tooling usage (**E5**), in addition to identity controls?

What you'll need. Your organization's Essential Eight target maturity level per strategy, an accurate workload inventory, authoritative patching sources for OS and applications, and (where used) allowlisting or endpoint protection tooling that pairs with CSW for E1.

Where to start. Sections 1–4 for scope and sensor rollout; 5–6 for ADM baselines aligned to E1/E4/E5; 7 for maturity-level evidence; 8–10 for audit exports and boundaries.

CSW primer — if you are new to Cisco Secure Workload

Cisco Secure Workload (CSW) is a **workload protection platform**. A lightweight **agent** on each server/VM/container observes processes and network flows; **cloud connectors** add AWS/Azure/GCP inventory where agents are not deployed.

CSW term	Meaning	Why compliance teams care
Scope	Logical boundary (CDE, PHI zone, CUI enclave)	Defines the systems you must prove are isolated
Label / filter	Tag or query assigning workloads to scopes	Automates scope membership — reduces drift
ADM	Application Dependency Mapping from observed traffic	Live network/data-flow diagram for assessors
Workspace	Policy container for one scope or application	Where allow/deny rules are authored
Monitor → Simulation → Enforce	Safe rollout sequence	Simulation = change-board evidence before blocking traffic
Denied Connections	Log of flows blocked by policy	Primary proof that enforcement operates

Console areas: Investigate (inventory, flows, vulns) · Defend/Segmentation (policy) · Manage (agents) · Platform (connectors) · Administration (audit log)

Read next: [Compliance evidence playbook](#) (full step-by-step) · [About CSW](#) (platform intro)

Universal evidence workflow

Execute these phases for **this framework's** compliance boundary. Map exports to control IDs in the **Reporting & Evidence** section below.

Phase	Goal	Key CSW actions	Typical duration
1 — Coverage	Every in-scope workload in CSW	Install agents/connectors ; apply compliance labels; create scope	Days 1–10
2 — Baseline	Machine-generated flow map	Run ADM ≥2 weeks; export clusters; app-	Days 11–28

Phase	Goal	Key CSW actions	Typical duration
		owner signoff	
3 — Policy	Designed isolation before enforce	Build workspace; Simulation mode; fix false positives	Days 29–45
4 — Operate	Continuous proof between audits	Enforce; quarterly export pack; ADM refresh every 90 days	Ongoing

Phase 1 checklist (coverage)

- ☐ In-scope host list reconciled to CSW Inventory (100% or documented exceptions)
- ☐ Labels applied: `compliance:<framework>`, `data:<classification>`, `env:<tier>`
- ☐ CSW scope created matching compliance boundary
- ☐ **Export:** inventory CSV + agent status screenshot

Phase 2 checklist (baseline)

- ☐ ADM running on compliance scope for full business cycle (≥ 2 weeks)
- ☐ Unexpected flows documented (shadow IT, vendor egress, scope creep)
- ☐ App owners signed cluster-to-application mapping
- ☐ **Export:** ADM diagram + flow samples with process context

Phase 3 checklist (policy)

- ☐ Default-deny posture defined for sensitive scope
- ☐ ADM-imported rules refined; Simulation run ≥ 1 week
- ☐ Change tickets for false-positive fixes; exception register updated
- ☐ **Export:** policy export + simulation report

Phase 4 checklist (operate)

- ☐ Enforcement enabled on pilot scope; negative test recorded in Denied Connections
- ☐ Quarterly pack: inventory, policy, denies, vulns, audit log (see playbook)
- ☐ SIEM integration verified (sample events)
- ☐ **Export:** enforcement screenshot + quarterly binder

What CSW evidence does not replace

Physical access, HR/training records, encryption key management, signed BAAs/vendor contracts, formal pen tests, and assessor attestation still require separate programmes. CSW addresses the **workload-resident** slice: segmentation, flows, process context, vuln reachability, and change drift.

CSW effectiveness for this framework

- E2/E6 patch prioritisation via CVE+EPSS+reachability
- E5 admin privilege path restriction
- ML1–ML3 maturity evidence progression

Compared to manual programmes: static diagrams and annual firewall samples age immediately; CSW ties evidence to **live workload behaviour** and produces queryable exports on demand — supporting "operating effectively" language in PCI v4.0, SOC 2 CC7, and HIPAA risk analysis.

1. Overview

The **Essential Eight** is a prioritized set of mitigation strategies published by the Australian Cyber Security Centre (ACSC). The **Essential Eight Maturity Model (EEMM)** defines three maturity levels (**ML1, ML2, ML3**) with increasing rigor. **CSW is not a complete Essential Eight solution.** It **strongly complements** patch prioritization and admin-path containment (**E2, E5, E6**), and provides **meaningful but partial** visibility for application execution and hardening signals (**E1, E4**). Several strategies are **predominantly out of CSW scope** (**E3** configuration, **E7** MFA, **E8** backups), though CSW may still provide **peripheral** telemetry (for example, flows associated with backup or macro-enabled application behavior).

1.1 Essential Eight Strategies and CSW Support Summary

Strategy	Name	CSW support level	Primary CSW mechanisms
E1	Application Control	Partial / complementary	Process and hash visibility; execution telemetry; forensic hunting for unapproved binaries <i>alongside</i> allowlisting
E2	Patch Applications	Strong	Software inventory, CVE awareness, EPSS where available, reachability/exposure prioritization
E3	Configure Microsoft Office Macro Settings	Limited	Detection of macro-enabled processes with unexpected network egress

Strategy	Name	CSW support level	Primary CSW mechanisms
			<i>after the fact</i> ; does not set Office trust settings
E4	User Application Hardening	Partial	Unexpected connection patterns from browsers/office suites; depends on baseline/normalcy
E5	Restrict Administrative Privileges	Strong (network path); partial (usage)	Microsegmentation of admin ports; observation of admin tool usage at process level
E6	Patch Operating Systems	Strong	OS inventory, CVE mapping, prioritization via exposure
E7	Multi-Factor Authentication	Out of scope	Identity provider / IdP function; CSW does not perform MFA
E8	Regular Backups	Out of scope (core)	Optional: monitor backup traffic paths; does not prove backup integrity or immutability

1.2 How to Read This Reference design with the EEMM

The ACSC defines **maturity levels per strategy**, not globally. Your evidence pack should **label each strategy separately**. CSW evidence is typically **cross-cutting** (inventory, flows, policy enforcement) and must be **mapped** to the subset of EEMM requirements it can support—never substitute CSW for endpoint configuration or IdP controls.

2. Pre-Deployment Checklist

- ☐ Target maturity (**ML1 / ML2 / ML3**) documented **per strategy** with accountable owners
- ☐ CSW cluster available; connectivity from in-scope workloads (typically TCP 443 outbound)

- ❑ Sensor support validated for Windows/Linux versions in scope (patch posture **E6** prerequisites)
 - ❑ Cloud connectors configured where workloads run on AWS, Azure, or GCP
 - ❑ Change windows approved for agent rollout; communications plan for security and application teams
 - ❑ Complementary tools identified: **E1** allowlisting, **E3** Office GPO/MDM, **E7** IdP, **E8** backup/DR
 - ❑ Evidence storage location defined (retention, classification, access control) for audit exports
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3. Phase 1 — Sensor Deployment (Days 1–5)

3.1 Install Sensors (Linux)

```
sudo rpm -ivh tet-sensor-<version>.rpm      # RHEL/CentOS/Rocky/Alma
sudo dpkg -i tet-sensor-<version>.deb        # Debian/Ubuntu
```

```
sudo systemctl enable --now csw-agent 2>/dev/null || sudo systemctl
enable --now tetration-agent
systemctl status csw-agent || systemctl status tetration-agent
```

3.2 Install Sensors (Windows)

Use the CSW-provided Windows installer package from your tenant; verify service health:

```
Get-Service | Where-Object { $_.Name -match 'tetration|csw' }
```

3.3 Initial Posture

- **Enforcement:** Monitoring Only until ADM and owner sign-off (**E2/E6** patching still proceed via patch management; CSW does not patch)
 - **Tagging:** Use labels that support Essential Eight reporting, for example e8:e1, e8:e5-admin, tier:crown-jewel, patch_group:windows-servers
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4. Phase 2 — Scope Design for Essential Eight

4.1 Recommended Scope Layout

Enterprise

└─ Crown-Jewels-Apps	
└─ Windows-Servers	# E6 heavy
└─ Windows-Desktops	# E1/E3/E4/E5
└─ Linux-Servers	# E2/E6 heavy
└─ Admins-Jump-Zone	# E5 path controls
└─ Backup-Infrastructure	# E8 adjacent telemetry only

4.2 Label Taxonomy

Label	Values	Purpose
e8_strategy	E1...E8	Filter evidence by strategy narrative
admin_tier	tier0/tier1/user	E5 path policies
internet_exposed	true/false	E2/E6 prioritization
patch_source	wsus, sccm, vendor	Correlate inventory to patch authority

5. Phase 3 — Visibility and Baseline (ADM)

5.1 ADM for Unexpected Behavior (**E1, E4, E5**)

CSW UI → Investigate → Application Dependency Mapping

→ Workspace: "EssentialEight-Baseline"

→ Scope: Windows-Desktops + Linux-Servers (iterate)

→ Duration: ≥ 14 days (capture patch Tuesdays and admin windows)

Baseline questions:

Question	Strategy
Which executables initiate network connections on endpoints?	E1, E4
Do browsers talk to rare external IPs or ports?	E4
Which hosts initiate inbound admin protocol sessions?	E5
Do Office processes spawn child processes with egress?	E3 (complementary)

5.2 Software Inventory for Patch Posture (**E2, E6**)

CSW UI → Investigate → Inventory

→ Filter: OS family, package/version, last observed

→ Export CSV weekly for reconciliation with WSUS/SCCM/Intune or Linux repos

Use inventory **gaps** (unknown agents) as ML2/ML3 coverage metrics for **visibility completeness**, not as a substitute for patch SLAs.

6. Phase 4 — Policy Design and Enforcement

6.1 E5 — Administrative Privilege Paths (Illustrative Policies)

CSW enforces **who can reach** admin services—not **who authenticates**. Pair with PAM/MFA (E7).

DENY: Any-NonJumpHost → Windows-Servers (dst ports 3389, 5985, 5986)

DENY: Any-NonJumpHost → Linux-Servers (dst port 22) except Bastion-Subnet

ALLOW: Admins-Jump-Zone → Windows-Servers (3389/RDP gateway paths as validated)

ALLOW: Admins-Jump-Zone → Linux-Servers (22)

LOG: any other administrative port attempts

6.2 E2 / E6 — Network Containment While Patching (Compensating)

During delayed patching on exposed tiers, **narrow** connectivity:

DENY: Internet → Vulnerable-App-Tier (except approved WAF/load-balancer paths)

ALLOW: Patch-Management-Infra → Windows-Servers / Linux-Servers (update ports per tool)

6.3 E1 / E4 — Detection-Oriented Rules (Complementary)

After baseline, alert on **first-seen unsigned binary path** or **unexpected child of browser**—tune aggressively to avoid noise:

ALERT: first_seen_process = true AND scope = Windows-Desktops

ALERT: parent_process in (browser_set) AND dst_port not in (443, 80)
AND dst not in approved_proxy

Application allowlisting (E1 primary) remains the authoritative control; CSW **augments** with network/process context.

6.4 Workspace Publication

CSW UI → Defend → Segmentation

→ Workspace: "EssentialEight-E5-AdminPaths"

→ Mode: Simulation (2+ weeks) → staged Enforcement

7. Phase 5 — Maturity Level Mapping (ML1–ML3)

The ACSC defines requirements per strategy at each maturity level. The tables below state **what CSW evidence commonly supports**—your assessor’s official checklist remains authoritative.

7.1 E1 — Application Control

Maturity	ACSC intent (summary)	CSW evidence that may support	CSW limitations
ML1	Basic controls for workstations	Process inventory and notable execution telemetry samples	Not an allowlist enforcer
ML2	Stronger coverage/focus areas	Longer ADM baselines; alerts on rare executables with network activity	Tuning burden; requires baselining
ML3	Organization-wide / advanced	Coverage dashboards (% agents active per desktop population); hunt exports	Does not replace centralized application control policy

7.2 E2 — Patch Applications & E6 — Patch Operating Systems

Maturity	CSW evidence that may support	Notes
ML1	Inventory exports showing versions; CVE lists for in-scope apps/OS	Prioritization narratives for risk acceptance
ML2	Reachability-informed priority queues; evidence of exposed vs internal-only risk	Pair with patch ticket closure from patch tool
ML3	Continuous monitoring of inventory drift; alerting on high-risk deltas	CSW detects drift; patching action is external

7.3 E3 — Office Macros

Maturity	CSW evidence that may support	CSW limitations
ML1–ML3	Flows showing WINWORD/EXCEL/POWERPNT child processes with egress	Does not prove macro trust settings or ASR rules

7.4 E4 — User Application Hardening

Maturity	CSW evidence that may support	CSW limitations
ML1	Basic external connection inventory from browsers	No browser configuration management
ML2–ML3	Anomaly alerts on rare destinations, protocols, or lateral use	Requires sound baseline; false positives possible

7.5 E5 — Restrict Administrative Privileges

Maturity	CSW evidence that may support	Pair with
ML1	Visibility into admin protocol usage	Jump host/PAM design
ML2	Simulation and then enforcement restricting admin ports	IdP + MFA (E7) for actual privilege restriction
ML3	Ongoing conformance dashboards; deny logs for policy violations	Privileged access reviews (process outside CSW)

7.6 E7 — Multi-Factor Authentication

Maturity	CSW role
ML1–ML3	No primary CSW mapping. Maintain IdP/MFA evidence separately. Optional: enforce paths to auth infrastructure only.

7.7 E8 — Regular Backups

Maturity	CSW role
ML1–ML3	No backup immutability or restore testing. Optional: monitor backup client flows; alert on backup traffic to unexpected destinations.

8. Phase 6 — Control-by-Control Mapping Table

Essential Eight requirement (strategy)	CSW capability	Evidence produced
E1 Application control	Process visibility; execution telemetry;	Process export for sample period; alerts on

Essential Eight requirement (strategy)	CSW capability	Evidence produced
	forensic search	anomalous executions <i>with</i> allowlist policy records from endpoint tool
E2 Patch applications	Inventory; CVE + EPSS + exposure	Vulnerability CSV; prioritized remediation list with reachability column
E3 Office macros	Egress patterns from Office processes	Flow exports for investigation windows; not macro policy config
E4 Application hardening	Browser/Office network baselines; anomalies	ADM diff; alert history
E5 Restrict admin privileges	Segmentation around admin services; process visibility	Policy exports; deny logs; jump-host path diagrams backed by flows
E6 Patch operating systems	OS inventory; CVE mapping	Same as E2 , OS-focused filters
E7 Multi-factor authentication	N/A (identity)	Maintain MFA platform logs; CSW optional path allowlists only
E8 Regular backups	Peripheral flow monitoring	Backup traffic baselines; anomaly alerts plus backup product reports

9. Boundaries — What CSW Does **Not** Cover

- **Application allowlisting** implementation (**E1** primary control) and **Microsoft Office macro trust configuration** (**E3**)
- **Multi-factor authentication** enrollment and enforcement (**E7**)
- **Backup** scheduling, immutability, offline copies, and restore testing (**E8**)
- **Endpoint hardening baselines** (ASR, AppLocker, browser settings) for **E4**
- **Patch deployment** itself (**E2/E6**): CSW informs priority; patch systems execute change
- **People and process** maturity artifacts (policies, training records, governance minutes) required by EEMM assessments

10. Audit Preparation and Evidence Export

10.1 Suggested Evidence Schedule

Evidence	CSW source	Suggested frequency	Strategies served
Agent coverage by population	Manage → Agents	Monthly	E1–E6 (visibility completeness)
ADM baseline snapshot	Investigate → ADM	90 days	E1, E4, E5
Inventory export (OS + apps)	Investigate → Inventory	Weekly	E2, E6
Vulnerability prioritized export	Investigate → Vulnerability	Monthly	E2, E6
Segmentation policy version	Defend → Segmentation	Each change	E5 (paths), compensating E2/E6
Alert sample pack	Alerts / SIEM	Quarterly or ad hoc	E1, E4, E5 incidents

10.2 Export Workflows

CSW UI → Investigate → Flow Search

- Scope + time window
- Export CSV/JSON with process metadata

CSW UI → Investigate → Process Search

- Filter: suspicious parent chains or first-seen binaries
- Export for IR or assessor sampling

CSW UI → Defend → Segmentation

- Export policy workspace and change history notes

10.3 Bash Helper — Evidence Folder Staging (Illustrative)

```
#!/usr/bin/env bash
set -euo pipefail
# Example output: 2026-Q2
AUDIT_CYCLE="$(date +%Y-Q$(( ( $(date +%m) - 1 ) / 3 + 1 )))"
TARGET_ROOT="/secure/evidence/acsc-e8/${AUDIT_CYCLE}"
mkdir -p "${TARGET_ROOT}/{inventory,vuln,policies,adm}"
echo "Copy CSW UI/API exports into the subfolders and generate MANIFEST.sha256"
find "${TARGET_ROOT}" -type f -print0 | sort -z | xargs -0 sha256sum > "${TARGET_ROOT}/MANIFEST.sha256"
```

Store manifests and exports under your records management policy; redact sensitive endpoints for third-party sharing.

11. Common Pitfalls

Pitfall	Mitigation
Claiming CSW satisfies E7 or E8	Keep IdP and backup evidence separate; use CSW only as supplementary
Over-alerting on E4 without baseline	Run ADM for full business cycles before enforcement/detection thresholds
E5 enforcement breaking emergency access	Document break-glass paths; alert-only exceptions with expiry
Inventory false confidence	Reconcile CSW inventory with CMDB; chase agents not reporting
Assuming macro security is “fixed” due to CSW	Validate Office trust settings via MDM/GPO and ASR

Appendix A — Related Frameworks

- [CIS Controls v8](#) — control themes aligned to hardening and visibility narratives.
- [NIST SP 800-207](#) — zero trust segmentation patterns supporting **E5** path restriction.
- [ISO/IEC 27001:2022](#) — for organizations mapping Essential Eight to broader ISMS audits.
- [UK NCSC CAF v3.2](#) — E2 and E6 patch evidence is the reuse path for CAF B4.

Appendix B — Official References (External)

- Australian Cyber Security Centre — *Essential Eight* and *Essential Eight Maturity Model* publications: <https://www.cyber.gov.au/resources-business-and-government/essential-cyber-security/essential-eight>
Verify you use the **current** ACSC version and assessment guidance; maturity requirements evolve.

Document prepared for Cisco customer engagements. Tailor scopes, maturity targets, and evidence indexes to your ACSC assessment boundary before external sharing.